Simon Chen

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Education

Boston University Boston, MA
Bachelor of Arts in Computer Science *Graduated: Spring 2026*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases, Machine Learning, Data Science.

Experience

Software Engineering Intern February 2026 – June 2026
RESET Standard *Shanghai, China*

- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Redesigned project-device mapping workflows in response to client requests, refactoring legacy forms to streamline onboarding of environmental monitoring devices.

Lead Product Engineer December 2025 – Present
EZ Esports *New York, NY*

- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

Projects

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS* May 2026

- Replaced KDE inference with a GBM theo layer, cutting model query latency from ~1.5ms to ~20μs per query (~75x) behind an unchanged live serving interface.
- Cut backtest and parameter-sweep iteration from ~14 hours on laptop CPUs to ~15 minutes by moving tick backtests and sweeps to Modal NVIDIA GPU workers with FAISS GPU search and CuPy-vectorized math.
- Sped up 155K-row neighbor search from 758μs to 154μs p50 by caching FAISS IVF indexes and pinning FAISS threads to avoid OpenMP oversubscription.
- Built and operated a live Kalshi BTC prediction-market trading system end to end (market-data ingestion, theo serving, strategy evaluation, execution, and risk), generating $8K net PnL on $200K traded volume in its first 4 days.

Food Waste Platform (BU Spark!) | *Django, JS, Auth0* September 2024 – December 2024

- Integrated Google Maps JavaScript API for interactive event discovery with user geolocation and distance calculations, implementing multi-criteria filtering across 18 food categories and 8 allergen types using Django class-based views with form validation
- Developed food waste reduction platform connecting students with leftover event food, implementing dual authentication (Django + Auth0 OAuth), QR code email delivery system, and many-to-many reservation management with real-time capacity tracking
- Deployed production Django application with WhiteNoise static file serving and automated Gmail SMTP notifications featuring inline QR code attachments, managing SQLite database with 100+ campus events while implementing comprehensive security features including CSRF protection and input validation

Technical Skills

Languages: Python, SQL, C++17, JavaScript, TypeScript, Java, Ruby
Frameworks & Libraries: Django, React, FAISS, CuPy, FastAPI, Vite, Next.js, Tailwind CSS
Databases & Infrastructure: PostgreSQL, SQLite, Redis, Modal, Docker, Nginx, Cloudflare